# Teorie pole

## Princip extremální akce

- prostor historií $\mathcal{H}$
  - "trajektorie"
- funkcionál akce $S: \mathcal{H} \to \mathbb{R}$
  - na konečném čase dává konečnou hodnotu
- klasické řešení (nekvantová) hledá ty správné historie
  → jsou dány podmínkou $\delta S[h] = 0$ při fixovaných o.p.
- typicky, ale ne nutně se historie volí trajektorií $x^a(t)$

## Lagrangeovský formalismus

- skutečné trajektorie

$$S[x(t)] = \int_{t_2}^{t_1} L(x^a(t), \dot{x}^a(t), t)\, dt$$

může

argumenty jsou obecně nezávislé

$L(x^a, v^a)$

- Lagrangián – $x^a$ ... polohy
  $v^a$ ... rychlosti

⇒ pohybové rovnice

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v}(x,\dot x)\right) - \frac{\partial L}{\partial x}(x,\dot x) = 0$$

## Hamiltonovský formalismus

- fázový prostor $[x^a, p_a]$
- pozorovatelné $F(x^a, p_a)$
- polohy $x^a$ nemusí mít vektorovou strukturu
- vždy platí symplektická struktura daná Poissonovými závorkami:

$$\{F,G\} = \sum_n \left(\frac{\partial F}{\partial x^a}\frac{\partial G}{\partial p_a} - \frac{\partial F}{\partial p_a}\frac{\partial G}{\partial x^a}\right)$$

- speciální pozorovatelná – Hamiltonián $H(x^a, p_a)$, platí:

$$\dot F = \{F, H\}$$

$$\dot x^a \to \dot x^a = \{x^a, H\} = \frac{\partial H}{\partial p_a}$$

$$\dot p_a \to \dot p_a = \{p_a, H\} = -\frac{\partial H}{\partial x^a}$$

- souvislost s Lagrangeem:
  $x, v \leftrightarrow x, p$ $\qquad L \leftrightarrow H$
  pro nedegenerovaný problém
- Legendreova transformace

$p = p(x,v)$ $\qquad p_a(x,v) = \frac{\partial L}{\partial v^a}(x,v)$

$v = v(x,p) =$

Chceme: $v(x, p(x,v)) = v$ $\qquad p(x, v(x,p)) = p$

$$\Rightarrow \quad H(x,p) = \sum_a p_a v^a(x,p) - L(x, v(x,p))$$

lze otočit

$$L(x,v) = \sum_a p_a(x,v)\, v^a - H(x, p(x,v))$$

## Polní teorie – funkcionální prostory

- $M$ prostor funkcí na $M$ : $\mathcal{F}M$
- $\mathcal{F}M$ vektorový prostor + topologické vlastnosti
- $\varphi \in \mathcal{F}M$ $\qquad \varphi(x) \in \mathbb{R}$ $\qquad x \in M$
  → $\varphi^x := \varphi(x)$ je komponenta vektoru
- $\omega \in (\mathcal{F}M)^*$ $\qquad \omega_x$

$$\langle \omega, \varphi\rangle = \sum_x \omega_x \varphi^x = \int \boxed{\omega_x}\boxed{\varphi^x}\boxed{dV}$$

ve skutečnosti $\omega_x dV$ je hustota
→ diferenciál Lebesgueovy míry

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## Parciální derivace na $\mathbb{R}^n$

- prostor bodů $A$
- prostor směrů $\Delta\vec x$
- $f$ funkce na bodech $f(A)$

⇒ změna $f$ při posunu

$$f(x_0 + \varepsilon\Delta\vec x) = f(x_0) + \varepsilon\, df(x_0) + O(\varepsilon^2)$$

$$df(x_0) = \frac{d}{d\varepsilon} f(x_0 + \varepsilon\Delta\vec x)\Big|_{\varepsilon=0} = \sum_n \underbrace{\frac{\partial f}{\partial x^a}(x_0)}_{\text{parciální derivace}}\Delta x^a$$

## Variace = ∂ v nekonečné dimenzi

- prostor historií /funkcí ... $H$, $\varphi, \psi, \dots$
- směry na $H$ ... $\delta\varphi, \delta\psi$
- funkcionály na $H$ ... $F[\varphi]$

⇒ změna $F$ při posunu

$$F[\varphi_0 + \varepsilon\delta\varphi] = F[\varphi_0] + \varepsilon\,\delta F[\varphi_0] + O(\varepsilon^2)$$

$$\delta F[\varphi_0] = \frac{d}{d\varepsilon}F[\varphi_0 + \varepsilon\delta\varphi_0]\Big|_{\varepsilon=0} = \int_x \underbrace{\frac{\delta F}{\delta\varphi^x}(\varphi_0)}_{\text{variace}}\delta\varphi^x\, dV$$

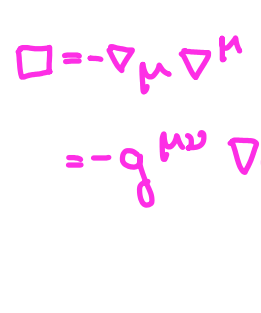
## Skalární pole

$\psi: M \to \mathbb{R}$ (prostoročas) $\qquad g = -dt^2 + dx^2 + dy^2 + dz^2$

$\Omega = \langle\Sigma_1|\Sigma_2\rangle$

akce

$$S_\Omega[\psi] = -\int_\Omega\left(\frac12\nabla\psi^2 + V(\psi)\right)d\Omega$$



$\nabla_\mu$ prostoročasový gradient

$V(\psi) = \frac12 m^2\psi^2 + \frac{1}{4!}\lambda\psi^4$ ? } není vidět, co se tu děje

$$S[\psi_0 + \varepsilon\psi] = -\int_\Omega\left(\frac12(\nabla\psi_0)^2 + V(\psi_0)\right)d\Omega - \varepsilon\int_\Omega\left(\nabla\psi_0\cdot\nabla\delta\psi + V'(\psi_0)\delta\psi\right)d\Omega + O(\varepsilon^2)$$

$$\delta S[\psi_0] = -\int_{\Omega}\left(\nabla\psi_0\cdot\nabla\delta\psi + V'(\psi_0)\delta\psi\right)d^4\Omega$$

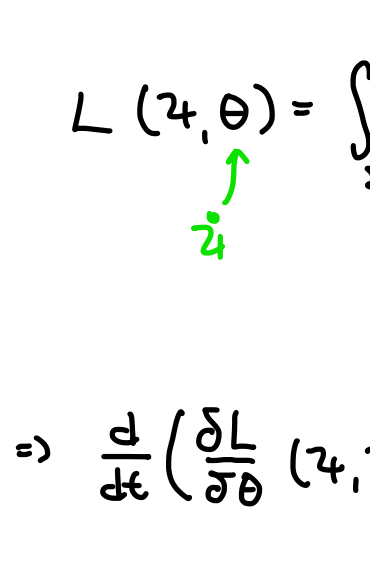
$$= -\int_\Omega\left(\nabla\cdot(\nabla\psi_0\delta\psi) - (\nabla\cdot\nabla\psi_0)\delta\psi + V'(\psi_0)\delta\psi\right)d^4\Omega$$

$$= -\underbrace{\int_{\partial\Omega}\nabla\psi_0\delta\psi\, d^3\Sigma}_{=0} - \int_\Omega\left(\Box\psi_0 + V'(\psi_0)\right)\delta\psi\, d^4\Omega$$

$\delta\psi|_\Sigma = 0$ $\qquad \Box = -\nabla_\mu\nabla^\mu = -g^{\mu\nu}\nabla_\mu\nabla_\nu$

$$\Rightarrow \quad \frac{\delta S_\Omega}{\delta\psi}[\psi_0] = -\Box\psi_0 - V'(\psi_0) = 0$$

pro $V = \frac12 m^2\psi^2$ ⇒ Klein-Gordonova rovnice



$\psi(x) = \varphi(t,x) = \varphi^x(t)$

$x = [t, x]$

$$\nabla\psi = dt\frac{\partial\psi}{\partial t} + \vec\nabla\psi$$

$$= dt\,\dot\varphi + \vec\nabla\varphi$$

$$(\nabla\psi)^2 = -\dot\psi^2 + (\vec\nabla\psi)^2$$

$$S[\psi(x)] = \int_{t_2}^{t_1}\int_{\Sigma_t}\left(\frac12\dot\psi^2 - \frac12(\vec\nabla\psi)^2 - V(\psi)\right)\underbrace{dV\,dt}_{d\Omega}$$

$$S[\psi(t)] = \int_{t_2}^{t_1} L(\psi(t), \dot\psi(t))\,dt$$

$$L(\psi,\underset{\dot\psi}{\theta}) = \int_{\Sigma_t}\left(\underbrace{\frac12\theta^2}_{\text{kinetický člen}} - \underbrace{\frac12\vec\nabla\psi^2 - V(\psi)}_{\text{potenciální člen}}\right)dV$$

$$\Rightarrow \quad \frac{d}{dt}\left(\frac{\delta L}{\delta\theta}(\psi,\dot\psi)\right) - \frac{\delta L}{\delta\psi}(\psi,\dot\psi) = 0$$

$\theta \to \theta + \varepsilon\delta\theta$ $\qquad \psi\to\psi$

$$\delta L = \int_{\Sigma_t}\theta\,\delta\theta\, dV \Rightarrow \frac{\delta L}{\delta\theta}(\psi,\theta) = \theta$$

$\theta\to\theta$ $\qquad \psi\to\psi+\varepsilon\delta\psi$

$$\delta L = \int_{\Sigma_t}\left(-\vec\nabla\psi\cdot\vec\nabla\delta\psi - V'(\psi)\delta\psi\right)dV$$

$$\overset{PP}{=} \int_{\Sigma_t}\left(\vec\nabla\cdot\vec\nabla\psi - V'(\psi)\right)\delta\psi\, dV$$

$$\Rightarrow \quad \frac{\delta L}{\delta\psi}(\psi,\theta) = \Delta\psi - V'(\psi)$$

$$\Rightarrow \quad \frac{d}{dt}(\dot\psi) - \Delta\psi + V'(\psi) = 0$$

$$\ddot\psi - \Delta\psi + V'(\psi) = 0$$

$$\Box\psi + V'(\psi) = 0$$

$$\pi(\psi,\theta) = \frac{\delta L}{\delta\theta}(\psi,\theta) = \theta$$

$\theta = \pi$

$$H(\psi,\pi) = \int_{\Sigma_t}\theta(\psi,\pi)\,\pi\, dV - L(\psi,\theta(\psi,\pi))$$

$$= \int_\Sigma\left(\pi^2 - \frac12\pi^2 + \frac12(\vec\nabla\psi)^2 + V(\psi)\right)dV$$

$$= \int_\Sigma\left(\frac12\pi^2 + \frac12(\vec\nabla\psi)^2 + V(\psi)\right)dV$$

$$\dot\psi = \frac{\delta H}{\delta\pi} = \pi \qquad \dot\pi = -\frac{\delta H}{\delta\psi} = \Delta\psi - V'(\psi)$$

$$\delta H = \int\left(\nabla\psi\cdot\nabla\delta\psi + V'(\psi)\delta\psi\right)dV$$

$$= \int\left(-\Delta\psi + V'(\psi)\right)\delta\psi\, dV$$

$$\to \quad \ddot\psi - \Delta\psi + V'(\psi) = 0$$

$$\Rightarrow \quad \Box\psi + V'(\psi) = 0$$

$$\{\psi^x, \pi_y\} = \delta^x_y = \delta(x|y)$$

$$\{F,G\} = \int_\Sigma\left(\frac{\delta F}{\delta\psi}\frac{\delta G}{\delta\pi} - \frac{\delta F}{\delta\pi}\frac{\delta G}{\delta\psi}\right)dV$$

funkce od $x$

## Akce EM pole

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$$

$$S_{EM}[A] = -\frac{\varepsilon_0}{4}\int_\Omega F_{\alpha\beta}F^{\alpha\beta}\, d^4\Omega$$

zadaný externí zdroj

lineární interakce s potenciálem

$S_{ext}[A]$ $\qquad$ Př.: $S_{ext}[A] = \int_\Omega J^\mu_{ext} A_\mu\, d^4\Omega$

$$\frac{\delta S}{\delta A_\alpha} = -\varepsilon_0\nabla_\mu F^{\alpha\mu} + \frac{\delta S_{ext}}{\delta A_\alpha}$$

$$= -\varepsilon_0\nabla_\mu F^{\alpha\mu} + J^\alpha_{ext} \overset{?}{=} 0$$

$$\Rightarrow \quad \nabla_\mu F^{\alpha\mu} = \frac{1}{\varepsilon_0}J^\alpha_{ext}$$

$$+ \quad \nabla_{[\gamma}F_{\alpha\beta]} = 0$$

## Lagrangeovský formalismus

$$A^\alpha = \begin{bmatrix}\phi\\ \vec A\end{bmatrix} \qquad F_{\mu\nu} = \begin{bmatrix}0 & -\vec E\\ \vec E & \vec B\end{bmatrix}$$

$$\vec E = -\vec\nabla\phi - \dot{\vec A} \qquad \vec B = \vec\nabla\times\vec A$$

$$-\frac14 F_{\mu\nu}F^{\mu\nu} = \frac12(E^2 - B^2)$$

$$S_{EM}[\phi,\vec A] = \int_{t_2}^{t_1}\int_{\Sigma_t}\frac{\varepsilon_0}{2}(E^2 - B^2)\,dV\,dt$$

externí pole musí splňovat rovnici kontinuity

$$0 = \nabla_\alpha J^\alpha \qquad J = \begin{bmatrix}\varrho\\ \vec j\end{bmatrix}$$

$$\Rightarrow \quad \dot\varrho + \vec\nabla\cdot\vec j = 0$$

$$L_{EM}(\phi,\vec A,\dot\phi,\dot{\vec A}) = \frac{\varepsilon_0}{2}\int_{\Sigma_t}(E^2 - B^2)\,dV$$

$$L_{ext}(\phi,\vec A) \qquad L_{ext} = \int_{\Sigma_t}\left(-\varrho_{ext}\phi + \vec j_{ext}\cdot\vec A\right)dV$$

$$\frac{d}{dt}\left(\frac{\delta L}{\delta\dot{\vec A}}\right) - \frac{\delta L}{\delta\vec A} = 0 \quad\wedge\quad \frac{d}{dt}\left(\frac{\delta L}{\delta\dot\phi}\right) - \frac{\delta L}{\delta\phi} = 0$$

$$\vec j = -\cancel{\frac{\delta L_{ext}}{\delta\dot{\vec A}}} + \frac{\delta L_{ext}}{\delta\vec A} \qquad \varrho = \cancel{\frac{\delta L_{ext}}{\delta\dot\phi}} - \frac{\delta L_{ext}}{\delta\phi}$$

$$\left(\frac{\delta L_{EM}}{\delta\dot{\vec A}}\right)^{\cdot} - \frac{\delta L_{EM}}{\delta\vec A} = \vec j \qquad \left(\frac{\delta L_{EM}}{\delta\dot\phi}\right)^{\cdot} - \frac{\delta L_{EM}}{\delta\phi} = -\varrho$$

$\dot{\vec A} \to \dot{\vec A} + \delta\dot{\vec A}$ $\qquad \delta L_{EM} = \varepsilon_0\int\vec E\cdot(-\delta\dot{\vec A})\,dV$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\dot{\vec A}} = -\varepsilon_0\vec E$$

$\vec A \to \vec A + \delta\vec A$ $\qquad \delta L_{EM} = -\varepsilon_0\int_\Sigma\vec B\cdot(\vec\nabla\times\delta\vec A)\,dV$

$$= \varepsilon_0\underbrace{\int_\Sigma\vec\nabla\cdot(\vec B\times\delta\vec A)\,dV}_{\text{na okraj} \Rightarrow 0} - \varepsilon_0\int_\Sigma(\vec\nabla\times\vec B)\cdot\delta\vec A\, dV$$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\vec A} = -\varepsilon_0\vec\nabla\times\vec B$$

$\delta\dot\phi = 0$ $\Rightarrow$ $\frac{\delta L_{EM}}{\delta\dot\phi} = 0$

$\delta\phi$ $\qquad \delta L_{EM} = \varepsilon_0\int_\Sigma\vec E\cdot(-\vec\nabla\delta\phi)\,dV$

$$= \varepsilon_0\int_\Sigma(\vec\nabla\cdot\vec E)\,\delta\phi\, dV$$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\phi} = \varepsilon_0\vec\nabla\cdot\vec E$$

$$\Rightarrow \quad -\dot{\vec E} + \vec\nabla\times\vec B = \frac{1}{\varepsilon_0}\vec j \qquad \vec\nabla\cdot\vec E = \frac{1}{\varepsilon_0}\varrho$$

$$\vec\nabla\cdot\vec B = 0 \qquad \vec\nabla\times\vec E = -\dot{\vec B}$$

## Hamiltonovský formalismus

$$L(\phi,\vec A,\dot\phi,\dot{\vec A}) = \frac{\varepsilon_0}{2}\int_\Sigma(E^2 + B^2)\,dV + \int_\Sigma\left(-\varrho_{ext}\phi + \vec j_{ext}\cdot\vec A\right)dV$$

$$\vec E = -\dot{\vec A} - \vec\nabla\phi \qquad \vec B = \vec\nabla\times\vec A$$

$$\vec\pi = \frac{\delta L}{\delta\dot{\vec A}} = -\varepsilon_0\vec E = \varepsilon_0(\dot{\vec A} + \vec\nabla\phi)$$

$\kappa = \frac{\delta L}{\delta\dot\phi} = 0$ $\qquad$ vazba – primární

$\kappa(x) = 0$ $\qquad \forall x\in\Sigma$

$K_\nu = 0$ $\qquad \forall\nu\in\mathcal{F}\Sigma$ $\qquad K_\nu = \int_\Sigma \kappa\nu\, dV$

zhlazení!

### Fázový prostor

$\phi, \vec A, \kappa, \vec\pi$

$$\{F_1,F_2\} = \int\left(\frac{\delta F_1}{\delta\phi}\frac{\delta F_2}{\delta\kappa} - \frac{\delta F_1}{\delta\kappa}\frac{\delta F_2}{\delta\phi} + \frac{\delta F_1}{\delta\vec A}\cdot\frac{\delta F_2}{\delta\vec\pi} - \frac{\delta F_1}{\delta\vec\pi}\cdot\frac{\delta F_2}{\delta\vec A}\right)dV$$

z nějakého důvodu je zajímavé:

$\{K_{\nu_1}, K_{\nu_2}\} = 0$
$\{\kappa(x), \kappa(y)\} = 0$ $\qquad x,y\in\Sigma$ } vazby první třídy

Jsou dimenzionálně odlišné od vazby druhé třídy
→ zde se vyskytují [illegible] [illegible] se synchronizují

$$\dot{\vec A} = \frac{1}{\varepsilon_0}\vec\pi + \vec\nabla\phi$$

$\dot\phi = \lambda$ → volné si libovolné, protože se tam nevyskytuje
⇒ něco jako Lagrangeův multiplikátor

$$H_\lambda = \int_\Sigma\left(\dot\phi\kappa + \dot{\vec A}\cdot\vec\pi\right)dV - L(\phi,\dot\phi,\vec A,\dot{\vec A})$$

$$= \int_\Sigma\left(\lambda\kappa + \left(\frac{1}{\varepsilon_0}\vec\pi - \vec\nabla\phi\right)\cdot\vec\pi - \frac{1}{2\varepsilon_0}\vec\pi^2 + \frac{\varepsilon_0}{2}\vec B^2 + \varrho_{ext}\phi - \vec j_{ext}\cdot\vec A\right)dV$$

$$= \int_\Sigma\left(\frac{1}{2\varepsilon_0}\vec\pi^2 + \frac{\varepsilon_0}{2}\vec B - \vec j_{ext}\cdot\vec A + \phi(\vec\nabla\cdot\vec\pi + \varrho_{ext}) + \lambda\kappa\right)dV$$

$$\frac{\delta H_\lambda}{\delta\phi} = \vec\nabla\cdot\vec\pi + \varrho_{ext} \qquad \frac{\delta H_\lambda}{\delta\vec A} = \varepsilon_0(\vec\nabla\times\vec B) - \vec j_{ext} = \dot{\vec\pi}$$

$$\frac{\delta H}{\delta\kappa} = \lambda = \dot\phi \qquad \frac{\delta H_\lambda}{\delta\vec\pi} = \frac{1}{\varepsilon_0}\vec\pi - \vec\nabla\phi = \dot{\vec A}$$

Pozorovatelné $F[\phi,\vec A,\kappa,\vec\pi; t]$

$$\dot F = \{F, H_\lambda\} + \frac{\partial F}{\partial t}$$

$$\dot K_\nu = \{K_\nu, H_\lambda\} + \frac{\partial K_\nu}{\partial t} \qquad K_\nu = \int\nu\kappa\, dV \qquad \nu = \nu(t)$$

$$= -\int_\Sigma\underbrace{\frac{\delta K_\nu}{\delta\kappa}}_{\nu}\frac{\delta H_\lambda}{\delta\phi}\,dV + K_{\dot\nu}$$

$$= -\int_\Sigma\nu(\vec\nabla\cdot\vec\pi + \varrho_{ext})\,dV + K_{\dot\nu} = 0$$

zachování vazby

$$\Rightarrow \quad G = \vec\nabla\cdot\vec\pi + \varrho_{ext} \equiv 0$$

$$G_\nu = \int_\Sigma\nu G\,dV$$

Gaussova vazba
sekundární

$$\Rightarrow \quad \dot K_\nu = G_\nu + K_{\dot\nu}$$

$$\dot G_\nu = \{G_\nu, H_\lambda\} + \frac{\partial G_\nu}{\partial t}$$

$$= -\int\frac{\delta G_\nu}{\delta\vec\pi}\cdot\frac{\delta H_\lambda}{\delta\vec A}\,dV + \int_\Sigma\left(\nu\dot\varrho_{ext} + \dot\nu(\vec\nabla\cdot\vec\pi + \varrho_{ext})\right)dV$$

$$= \;\vdots$$

$= G_{\dot\nu}$ $\qquad$ vymizí pro $G = 0$

algebra vazeb:

$$\{K_\mu, K_\nu\} = 0 \qquad \{G_\mu, G_\nu\} = 0 \qquad \{K_\mu, G_\nu\} = 0$$

$$\dot\phi = \frac{\delta H_\lambda}{\delta\kappa} = \lambda \qquad \dot\phi = \lambda$$

$$\dot\kappa = -\frac{\delta H_\lambda}{\delta\phi} = G \qquad \kappa = 0 \qquad \vec\nabla\cdot\vec E = \frac{1}{\varepsilon_0}\varrho_{ext}$$

$$\dot{\vec A} = \frac{\delta H_\lambda}{\delta\vec\pi} = \frac{1}{\varepsilon_0}\vec\pi - \vec\nabla\phi \qquad \vec E = -\dot{\vec A} - \vec\nabla\phi$$

$$\dot{\vec\pi} = -\frac{\delta H_\lambda}{\delta\vec A} = -\varepsilon_0(\vec\nabla\times\vec B) + \vec j_{ext} \qquad \vec\nabla\times\vec B - \dot{\vec E} = \frac{1}{\varepsilon_0}\vec j_{ext}$$

$$\vec\nabla\cdot\vec B = 0$$

$$H_\lambda = \int_\Sigma\left(\frac{1}{2\varepsilon_0}\vec\pi^2 + \frac{\varepsilon_0}{2}\vec B^2 - \vec j_{ext}\cdot\vec B\right) + G_\phi + K_\lambda$$

částečná redukce fázového prostoru

$$\phi = \int\lambda\, dt$$

$$\Rightarrow \quad \vec A, \vec\pi$$